

DRY-RUN GUIDE | 12_temperature.cpp
Source: cpp\1st_belt\12_temperature.cpp

CORE BEHAVIOR

Reads an integer temperature and selects a weather message from ascending threshold checks.

REPRESENTATIVE INPUT / CALL

Input: '25'.

STEP TRACE

25 is not below 0, 10, or 20; it is <=30, so that branch prints.

EXPECTED OUTPUT / RETURN

'Normal in temp'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Question:
003 | Write a program to read the temperature in centigrade and display a
004 | suitable message according to the temperature state below.
005 |
006 | Temp < 0: Freezing weather
007 | Temp 0-10: Very cold weather
008 | Temp 11-20: Cold weather
009 | Temp 21-30: Normal in temp
010 | Temp 31-40: It's hot
011 | Temp > 40: It's very hot
012 | */
013 |
014 | #include <iostream>
015 | using namespace std;
016 |
017 | int main() {
018 |     int temp;
019 |     cin >> temp;
020 |
021 |     if (temp < 0) {
022 |         cout << "Freezing weather";
023 |     } else if (temp <= 10) {
024 |         cout << "Very cold weather";
025 |     } else if (temp <= 20) {
026 |         cout << "Cold weather";
027 |     } else if (temp <= 30) {
028 |         cout << "Normal in temp";
029 |     } else if (temp <= 40) {
030 |         cout << "It's hot";
031 |     } else {
032 |         cout << "It's very hot";
033 |     }
034 |
035 |     return 0;
036 | }
037 |
```